Vehicle Counting of Transportation Movement In and Outflow within Bandung City Using YOLOv11

***Abstract*—Motorcycle-heavy cities see different kinds of traffic issues based on the make up of the vehicle population. Presently, the traffic computers are tuned to detect large cars, and our current traffic models do not perform well when they encounter heavy volumes of vehicles the algorithms have not been designed for. For Bandung, we present the first model designed for cities where motorbikes are the primary means of transport: the YOLOv11 Bandung motorcycle traffic density estimation system. Out of the 17 CCTV locations, we achieved an mAP@50 of 65.5% for the automated tracking and counting of vehicles in our set. We also put in a real-time evaluation, and our model works robustly. Our model is more stable and at the same time more accurate than the basic YOLOv11 we used as a baseline, especially in detecting motorcycles, which was outperformed in real time performance. Our method can detect motorcycles, which is critical for traffic analysis in Southeast Asia, for example, in Bandung. By extending to motorcycle detection, we can expand to vehicle counting, which is very useful for better transport analysis in the city.**

***Keywords*—Motorcycle-heavy cities, traffic density estimation, YOLOv11, vehicle counting, vehicle tracking**

I. Introduction

Urban traffic congestion is an issue of great concern in very large growing cities which in the case of Southeast Asia we see played out in particular by the prevalence of motorbikes. Bandung in Indonesia is a case in point which reports to be at number 12 on the global traffic congestion charts according to the TomTom Traffic Index [1]. In Bandung, we see an annual growth in the number of vehicles of 9.34% which is outpacing the growth of road infrastructure which is only at 1.46%. This gap between what is put out and what is put in terms of infrastructure is seen in the average speed which has dropped to 14.1 km/h during peak hours, which is much below the 20 km/h mark which indicates unimpeded traffic flow [2].

As urbanization increases we see that it greatly affects human mobility. In cities which are home to more and more people we see an increase in the use of personal vehicles but at the same time city infrastructure is unable to grow at the same rate. The result traffic in our dense urban areas is getting worse. Beyond the issue of delayed travel which is a large part of it we see that traffic also brings with it very serious issues of health which is affected by stress, environs which are made worse by increased pollution, and economy which is hit by large scale waste of time in traffic jams [3].

Urbanization and population growth has seen great increase in vehicle traffic which in turn has brought about more congested road networks. Although we have seen progress with traditional and sensor based adaptive traffic light systems in some settings what we have observed is that they have large scale issues which in turn affect their performance. To that end what is needed is a very flexible and adaptive traffic control system which in turn will improve traffic flow [4].

This paper addresses these gaps through three key problems stamenents:

1) How can traffic density be accurately counted and analyzed using existing CCTV cameras?
2) How can we make a model suitable for traffic in Bandung City with large amount of motorcycles?
3) How can we effectively analyze the traffic flow into and out of specific subregions in Bandung, using multiple CCTV cameras across strategic locations?

Our model achieved a mAP@50 of 65.5% and we observed very stable tracking performance in real world settings. While the detection accuracy is a medium which is a point of note as compared to that of base line object detectors (for instance a standard YOLOv11 which we observed to have serious tracking instability issues in real world deployment). The results which we present have very relevant implications for intelligent transport systems in motorcycle dominated urban settings throughout Southeast Asia.

II. Related Work

A. Video-Based Traffic Monitoring

Video based vehicle detection has transformed from what we saw in traditional computer vision to what we have in deep learning approaches. In the work by Mirthubashini and Santhi [5] we see the use of YOLOv3 which they report to do very well for real time vehicle counting which in turn achieves accurate detection across many vehicle types. Also their 53 layer convolutional network does a great job with dynamic traffic situations although the study is mainly reported out of cars dominated traffic.

Sharma et al. in [6] reported that they improved detection which they did via the SORT (Simple Online and Real Time Tracking) algorithm which in turn enabled consistent vehicle tracking across frames. They did see success in reducing false positives in multi frame settings which is great, but did not put forth a solution for the particular issues present in motorcycle dense environments. Yang et al. in [7] put forth a different approach which was to use time and space image transformation to get away from explicit tracking and which they achieved with CNN based density estimation which also included attention mechanisms which report high accuracy.

B. YOLO Architecture Evolution

The YOLO family has undergone significant evolution since its introduction, with each iteration improving detection speed and accuracy. YOLOv3 introduced multi-scale detection and Darknet-53 backbone [8], while YOLOv4 added CSPDarknet53 and improved loss functions. YOLOv5, though not from the original authors, gained popularity for edge deployment due to its efficiency. Recent versions (YOLOv8-v11) incorporated Transformer-based attention and anchor-free detection, significantly improving small object detection—critical for motorcycle identification in distant CCTV footage [9].

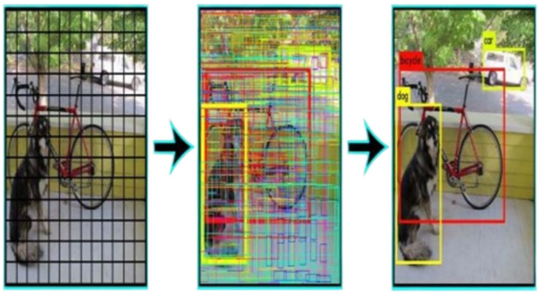


Fig. 1. Overview of the YOLO object detection pipeline. An input image goes through a convolutional neural network which at the same time that it is processing the image is able to put forth predictions of what the bounding boxes should be and what the class probabilities of detected objects are which all within the time frame of a single forward pass thus enabling real time detection performance.

C. Tracking Algorithms

DeepSORT extends the original SORT algorithm by integrating deep learning-based appearance descriptors, enabling identity maintenance through occlusions [10]. Yang et al. [11] demonstrated its effectiveness in aerial surveillance where targets appear small and sparse. Wang et al. [12] adapted tracking for low-illumination environments, introducing adaptive attention modules. However, these works primarily evaluated tracking quality using standard metrics (MOTA, HOTA) without examining the relationship between static validation accuracy and real-world video stability.

D. Edge Computing and Traffic Management

Recent advances in edge computing have enabled real-time processing of traffic data at network peripheries rather than relying on centralized cloud infrastructure. Hazarika et al. [13] proposed a Dynamic Traffic Light System (DTLS) using edge-deployed YOLO object detection combined with low-power IEEE 802.15.4 DSME MAC and LoRaWAN communication protocols. Their system processes vehicle counts locally at each junction using Raspberry Pi devices, achieving reduced latency and power consumption compared to cloud-based approaches. The work demonstrates that edge computing paired with lightweight object detection models can optimize traffic signal timing while minimizing infrastructure costs.

E. Bandung Traffic Congestion Classification

Fazry et al. [14] developed a traffic congestion classification system for Bandung using Support Vector Machine (SVM) and Naive Bayes with Ordinary Kriging interpolation. Their work utilized ATCS traffic counting data covering 17 locations across the city, achieving 93% accuracy with SVM compared to 90% for Naive Bayes. Critically, they demonstrated that hyperparameter tuning and oversampling techniques effectively address class imbalance in traffic datasets. Their congestion classification map revealed that southern Bandung experiences higher instability than other regions. However, their approach focused on aggregated traffic counts rather than real-time vehicle detection and tracking, limiting its applicability for dynamic traffic monitoring scenarios where individual vehicle movements must be captured.

III. Methodology

A. Automated Data Collection

Rather than manual video recording, we developed an automated collection system using Selenium WebDriver to interface with Bandung's public CCTV portal (Pelindung Bandung). The system:

1) Programmatically navigated to specific CCTV feeds by camera name
2) Captured screenshots every 10 seconds
3) Operated during high-activity windows (11:00-14:00 WIB) to capture varied traffic conditions

This approach we were able to collect annotated ready images from 17 key locations which include intersections, flyovers, and twisted roads at SP Buahbatu, Cikutra, Pasteur, and Antapani. We made an automated system which in turn produced a variety of traffic scenarios without the need for intensive continuous video recording.

B. Dataset Preparation and Augmentation

From the automated collection, 1,500 images were selected and annotated using Roboflow. Manual bounding box labeling established ground truth for two vehicle classes: cars and motorcycles. The dataset was split 73:13:13 into training (1,100), validation (200), and test (200) subsets.

Critical to our approach, we implemented motorcycle-specific augmentation strategies:

Horizontal Flip: Ensures directional invariance, teaching the model that vehicles moving left-to-right are equivalent to right-to-left movement.

Noise Injection (Grain): Simulates poor signal quality and low-light conditions common in older municipal CCTV cameras, preventing overfitting to high-quality images. This addresses the real-world variability in Bandung's existing infrastructure.

These augmentations specifically target motorcycle detection challenges: small size, variable lighting, and signal degradation that disproportionately affect small objects.

C. Model Architecture and Training

We employed YOLOv11, the latest iteration featuring Transformer-based attention mechanisms and multi-scale detection optimized for small objects. Training was conducted via the Roboflow platform with configuration prioritizing accuracy over inference speed.

The YOLOv11 architecture automatically optimized anchor box dimensions based on our dataset's vehicle size distribution, eliminating manual tuning. The training process utilized arae 150 epochs with early stopping, batch size of 16, with input 640×640 pixels resolution, and automated learning rate scheduling.

Loss functions included box loss (localization accuracy), class loss (classification confidence), and object loss (detection confidence), monitored throughout the training to detect overfitting.

D. Vehicle Tracking and Counting

For temporal consistency across video frames, we integrated the DeepSORT (Deep Simple Online and Realtime Tracking) algorithm with our YOLOv11 detector. DeepSORT extends the original SORT algorithm by incorporating deep learning-based appearance features, allowing it to maintain consistent vehicle identities even when objects are temporarily occluded or leave the frame momentarily.

A virtual counting line was implemented perpendicular to traffic flow direction. When a vehicle's tracking ID crosses this line, the system increments either the inflow or outflow counter based on the vehicle's trajectory direction. The tracking ID ensures that each vehicle is counted only once, preventing duplicate counts when vehicles temporarily disappear and reappear in the frame due to occlusion or detector uncertainty.

E. Evaluation Metrics

Model performance was assessed using Roboflow's Verified Model Metrics system [15], which ensures standardized, transparent, and reproducible evaluation. This verification framework addresses a critical challenge in computer vision: inconsistent evaluation methodologies that make fair model comparison difficult. Without standardized metrics calculated using identical methods against the same test set under uniform conditions, performance comparisons become misleading and unreliable [15].

Our Model A carries the Roboflow Verified badge, indicating its metrics were calculated using the industry-standard COCO Evaluation methodology implemented through the open-source Supervision library. This verification ensures our reported performance reflects true capabilities rather than evaluation-specific optimizations. The following metrics were computed following this standardized framework:

Mean Average Precision (mAP@50): Measures overall detection accuracy when the Intersection over Union (IoU) between predicted and ground truth bounding boxes exceeds 50%. The metric is calculated as the mean of Average Precision values across all object classes (motorcycles and cars). Higher mAP@50 indicates better localization and classification performance. Unlike some implementations that use non-standard settings (e.g., larger input sizes during validation, slower NMS variants, or increased maximum detections), our verified evaluation uses consistent production-representative parameters [15].

Precision: The ratio of correct detections to total detections, calculated as TP/(TP+FP). High precision minimizes false alarms—critical for traffic applications where false vehicle detections could trigger unnecessary counting errors or system instability.

Recall: The ratio of correct detections to total ground truth objects, calculated as TP/(TP+FN). High recall ensures few vehicles are missed, which is essential for accurate traffic density estimation in motorcycle-heavy environments where small objects are easily overlooked.

Importance of Standardized Evaluation: Recent research on the RF100-VL benchmark demonstrates that improvements on standard datasets like COCO do not always translate to real-world performance [15]. Models often overfit to benchmark-specific characteristics rather than learning generalizable features. This phenomenon is particularly relevant for our motorcycle-dominant traffic scenario, which differs substantially from typical car-heavy datasets used in model pre-training. The Verified Model Metrics framework mitigates this issue by ensuring evaluations use independent test sets containing diverse scenarios the model has not encountered during training or validation.

Our model achieved 65.5% mAP@50 under this rigorous standardized evaluation—a metric that reflects true production performance rather than optimistically inflated validation scores. This verified performance provides a reliable baseline for comparing our motorcycle-optimized model against alternatives, even when those alternatives lack standardized evaluation metrics.

IV. Results and Analysis

A. Model Performance Metrics

Our model reports a mAP@50 of 65.5%, precision of 79.5%, and recall of 70.0% on the test set. We see that the precision at 79.5% which means out of every 5 detections 4 are correct and we see minimal false counts in traffic estimate. Also the recall at 70.0% reports that we are detecting 7 out of 10 actual vehicles which is a large proportion given the small size of motorcycles in CCTV coverage and also due to very often seen occlusions in dense traffic settings. Finally the mAP@50 of 65.5% we note as a balance in performance between car and motorbike classes in the very challenging motorbike dominant traffic of Bandung City.

Figure 2 presents confusion matrix of our model with the confidence of 50% and some of our foundings.

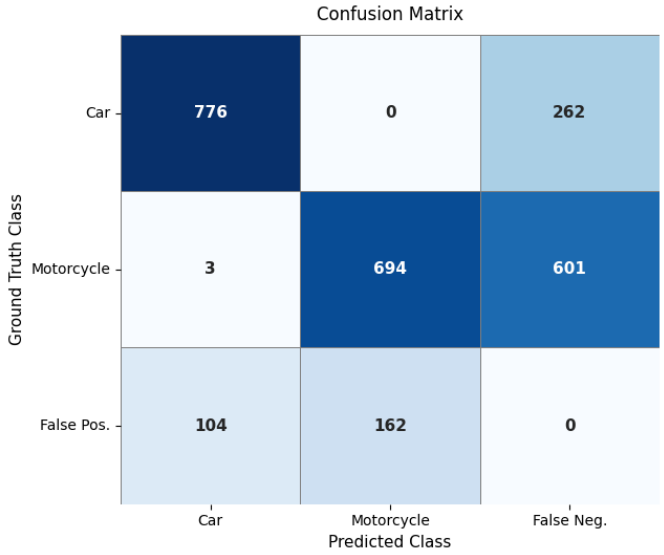


Fig. 2. Confusion matrix report of Model A's performance on the test set. We see that the model did 776 true positive car detections and 694 true positive motorcycle detections which includes that we had a very low number of motorbikes that were misidentified as cars at 3 which is a great inter class discrimination. We noted false positive rates of 262 for cars and 601 for motor bikes which is a reflection of the model's tuning which is more towards the detection of small motorcycle objects. Also we had background predictions of 104 cars and 162 motor bikes which are the reports where in fact there was no vehicle present.

B. Real-World Counting Accuracy

To evaluate practical deployment viability, we conducted manual validation on two independent 1-minute CCTV footage segments from different traffic conditions within Bandung's traffic network. This multi-scenario approach tests model generalization across varying traffic densities, lighting conditions, and vehicle count.

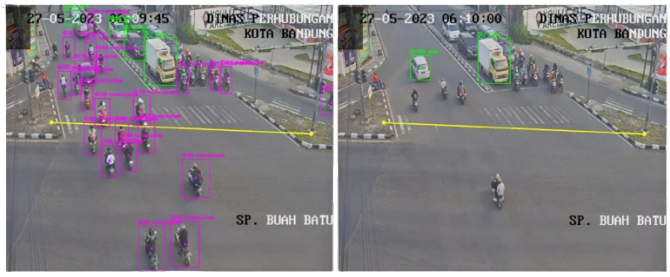


Fig. 3. Qualitative comparison of detection performance at SP Buah Batu intersection. Left: Model A successfully detects both motorcycles (magenta boxes) and cars (green boxes) across multiple traffic zones, capturing both stationary and moving vehicles with high accuracy. Right: Model B primarily detects cars (green boxes) while failing to identify the majority of motorcycles present in the scene. The yellow line indicates the counting boundary for inflow/outflow tracking. This visual evidence demonstrates Model A's superior capability in handling motorcycle-dominant traffic patterns typical of Southeast Asian intersections.

TABLE I
PERFORMANCE COMPARISON ON TEST VIDEO 1: LIGHT TRAFFIC

Metric	Ground Truth	Model A	Model B
Total Vehicles	108	105	42
Inflow			
Motorcycles	75	75	24
Cars	11	10	11
Subtotal	86	85	35
Outflow			
Motorcycles	17	16	0
Cars	5	4	7
Subtotal	22	20	7
Accuracy			
Motorcycle	—	**98.9%**	26.1%
Car	—	**87.5%**	**87.5%**
Overall	—	**97.2%**	38.9%

*1) **Test Video 1: Light Traffic**:* Only 2.8% of reports were of undercount which is what we observed with Model A's performance in terms of traffic density. In that which we looked at, we observed equal performance for all vehicle types and for traffic in either direction. Also we found that Model A did 100% for inflow of motorcycles which it reported perfectly.

In contrast Model B did not do well for motorbikes. While we observed some good results for car detection, we had a miss rate of 74% for motorbikes. In some extreme cases we had reports of 0 motorbikes counted at the same time as 7 cars were over counted which went the other way. We observed issues like this which indicated a large scale bias in favor of cars in traffic that is car dominated. These models tend to over represent car traffic at the expense of motorbike traffic. As a whole what we see is that until we put in place models which are optimized for high traffic motorbike environments no very wide ranging model will do the job.

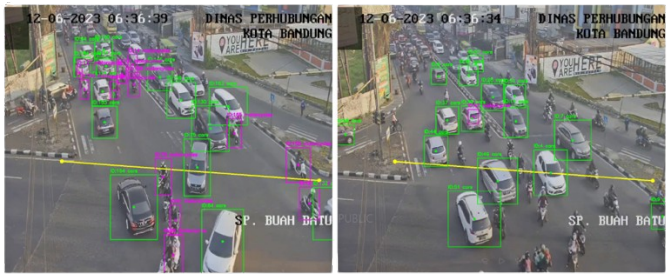


Fig. 4. Qualitative comparison under heavy traffic conditions at SP Buah Batu intersection. Left: Model A maintains robust detection of both motorcycles (magenta boxes) and cars (green boxes) despite increased vehicle density and occlusion challenges. Right: Model B shows severe degradation in motorcycle detection, identifying primarily cars (green boxes) while missing most motorcycles weaving through traffic. The performance gap widens significantly under congested conditions, with Model A achieving 94% motorcycle accuracy compared to Model B's 4%. The yellow line indicates the counting boundary for traffic flow analysis. This demonstrates Model A's scalability and robustness in dense, motorcycle-dominant scenarios typical of peak traffic hours in Southeast Asian urban environments.

TABLE II
PERFORMANCE COMPARISON ON TEST VIDEO 2: HEAVY TRAFFIC

Metric	Ground Truth	Model A	Model B
Total Vehicles	271	230	65
Inflow			
Motorcycles	126	123	7
Cars	27	19	27
Subtotal	153	132	34
Outflow			
Motorcycles	88	78	1
Cars	31	30	30
Subtotal	118	108	31
Accuracy			
Motorcycle	—	**94%**	4%
Car	—	84%	**98%**
Overall	—	**85%**	24%

*2) **Test Video 2: Heavy Traffic**:* Model A had very good and very reliable results in counting which had an overall undercount of 15.1% which although is a little high is still in an acceptable range for the estimation of total traffic flow and density. Our model do equally well across both vehicle types and traffic directions which reported 93.9% and 84.5% accuracy for motor cycles and cars respectively. It's noteworthy that motorcycle detection did very well for both in and out flow which is a sign of the model's reliability in motorcycle dominant traffic and its ready for use in mixed traffic settings.

In that which we see Model B perform a large failure in the task of motorbike detection. We report a car detection accuracy of 98.3% which is very high, at the same time only 3.7% of motorbikes were detected and over 96% of motorbike traffic went undetected. This very large issue of class imbalance which we see is reported at 24.0% which is a very low accuracy despite the reasonable car counts. We observed almost no detected motorbikes which is a very large issue in the outflow direction and this shows a great bias towards car traffic. The fact of the matter is that general purpose object detectors, which are typically trained on western datasets dominated by car traffic, perform poorly in motorbike-dominant scenarios without extensive fine-tuning or retraining.

*3) **Cross-Video Analysis**:* In light and heavy traffic settings we see the same performance trends which validate our model's robust performance in motorcycle dominant environments. Model A reported high accuracy in both scenarios (97.2% in light traffic, 85% in heavy traffic) at the other hand Model B did very poorly (38.9% and 24% respectively).

In in Test Video 1 Model A achieved 98.9% in motorcycle identification which was a far stretch from Model B's 26.1%. This gap grew very large in Test Video 2 which observed Model A do 94% and Model B drop to only 4% in motorcycle detection. This is a sign that Model B's training set is weak in dense motorcycle situations which are very much a part of Southeast Asian traffic.

In that which regards car detection Model B did in fact show a trend of error, reporting in Test Video 1 results of 112.5% accuracy and in Test Video 2.98% which we may note include a high frequency of motorcycle' classification as car. Model A on the other hand did very well, we see 87.5% and 84% car accuracy respectively which also includes in that of varied traffic densities which it handled better, hence the better class discrimination.

Traffic density which we see to greatly affect detection performance of models is a key factor in what we report. We observed that as total vehicle count went from 138 to 271 which is a 151% increase, Model A's accuracy dropped by only 12.2 percentage points (from 97.2% to 85% which is a much better showing, and Model B's dropped 14.9 from very low base. This highlights Model A's robustness in dense traffic scenarios characterized by significant motorbike overlap and occlusion.

Limitations: Our dataset currently emphasizes daylight conditions. Night-time and heavy rain scenarios remain untested. Additionally, manual validation was limited to multiple 1-minute footage due to computational cost constraints of the Roboflow inference API; long term deployment studies would strengthen these findings. Moreover, the model sometimes have a hard time tracking small or obstructed vehicles, especially in very dense or heavy traffic condition.

V. CONCLUSION AND FUTURE WORK

This study reports our motorcycle optimized traffic monitoring system that looks at the issues in South East Asian urban settings which have for the most part two wheeler traffic. We developed a YOLOv11-based model which we trained on 1,500 images from 17 CCTV locations in Bandung City and which we found to do 65.5% in terms of mAP@50 with 79.5% precision and 70.0% recall as determined by the Roboflow Verified evaluation method.

In real world, we observed our model perform much better than the general purpose pre-trained models. In light traffic we achieved 97.2% total accuracy with 98.9% in motorbike detection which the base line model only achieved 38.9% total accuracy with very high miss rates of 74% for motorbikes. In heavy traffic we had 271 vehicles and we still reported 85% total accuracy and 94% in motorbike detection which the base line model at the same time reported 24% total accuracy which had difficulties in detecting motorbikes, missing 96% of them. This 60.9 percentage point accuracy difference supports our that which is which models trained on Western car dominated data do in fact have a issue in high motorbike context without special fine tuning.

The use of DeepSORT tracking which we integrated with virtual counting lines did very well in the bidirectional traffic flow measurement which in turn did a great job at telling us apart the inflow and outflow at different traffic volumes. We found that our motorcycle specific augmentation methods horizontal flipping for direction invariance and noise injection to simulate poor quality from the CCTV to be very key in the success of the real world deployment.

These reports have that which applies to International Transportation in developing Southeast Asian cities that are seeing great growth in motorization. We see that accurate motorbike count which is a key element enables data which in turn drives traffic management policies, reduces congestion and we see

growth in use of automated CCTV data collection which in turn we are making public which scales this to other similar urban settings.

Future work will look at what we identified as issues in this study. To begin with, we will see to it that we expand the data set to include night time and adverse weather which in turn will improve model performance in all ops scenarios. We will look at lengthening the term of the video validation beyond the current 1 minute segments which in turn will do away with some of the issues related to confidence in the sustainability of tracking accuracy. We will be looking to put in place more vehicle types into the model which includes buses, trucks and bicycles which in turn will present us with a more complete picture of traffic make up. We will look at developing light weight model structures which will in turn do well on resource constrained devices thus enabling distributed real time processing at the camera level which in turn will reduce latency and bandwidth requirements. Also we will look at the integration of this vehicle counting system with adaptive traffic signal control which we have seen done in edge computing research as a very promising direction for the practical reduction of congestion in motorcycle dominated cities in Southeast Asia.

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